

Multimodal Large Language Models Are Not Calibrated



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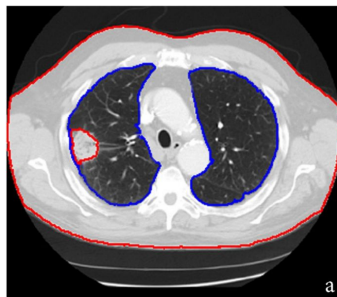


Zeynep
Akata

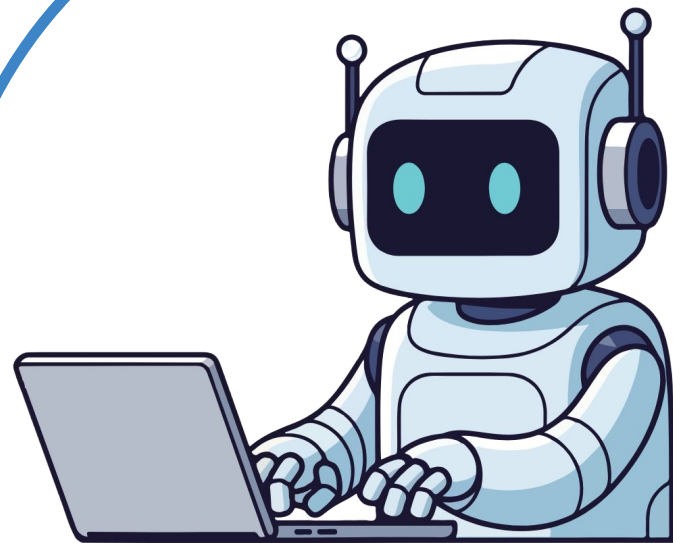
Paris, 24 September 2026

Motivation

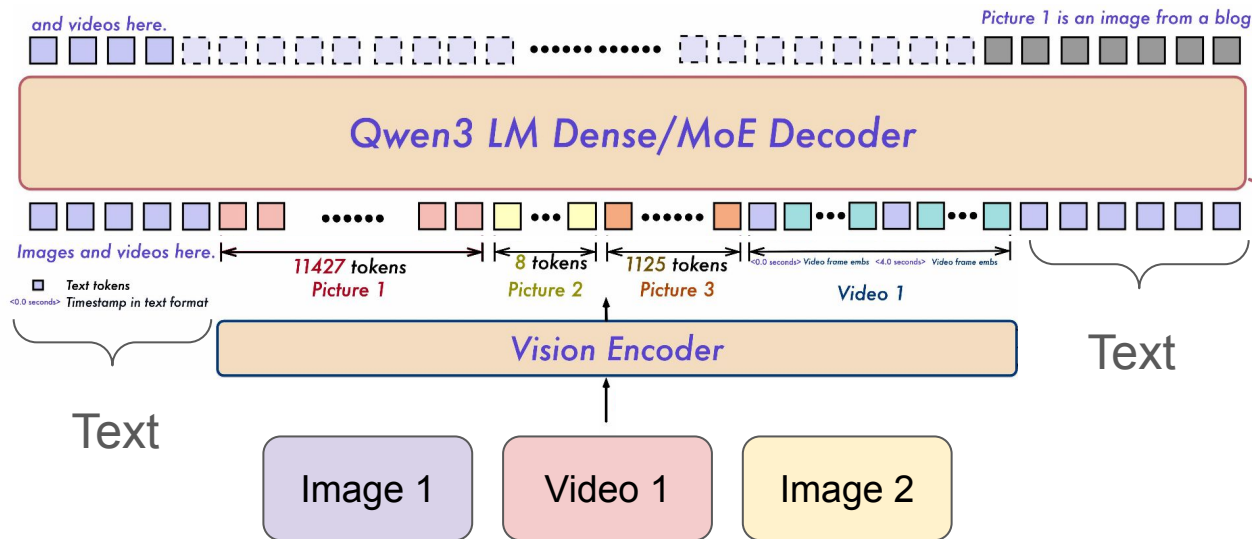
- Reliability in critical domains
- Track hallucinations
- Agentic domain



```
def gradient_descent(func, w0, gamma, max_iter):  
    w = w0.copy()  
  
    f_values = []  
    gradient_norms = []  
  
    for t in range(max_iter):  
        # COMPLETE HERE  
        w = w - gamma*grad_f(w)  
  
        f_values.append(func(w))  
        gradient_norms.append(np.linalg.norm(grad_f(w)))  
  
    return w, f_values, gradient_norms
```



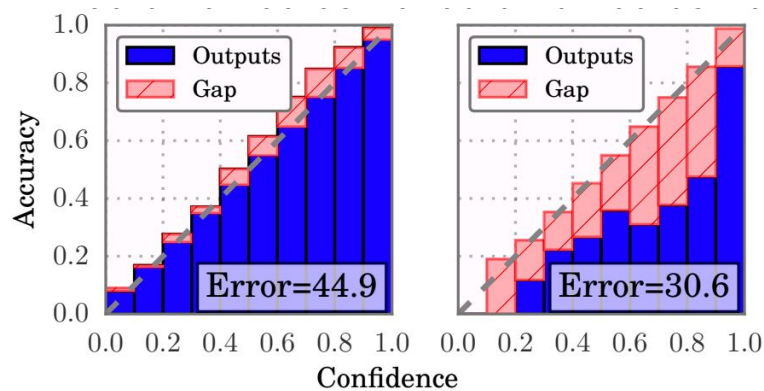
Multimodal Large Language Models



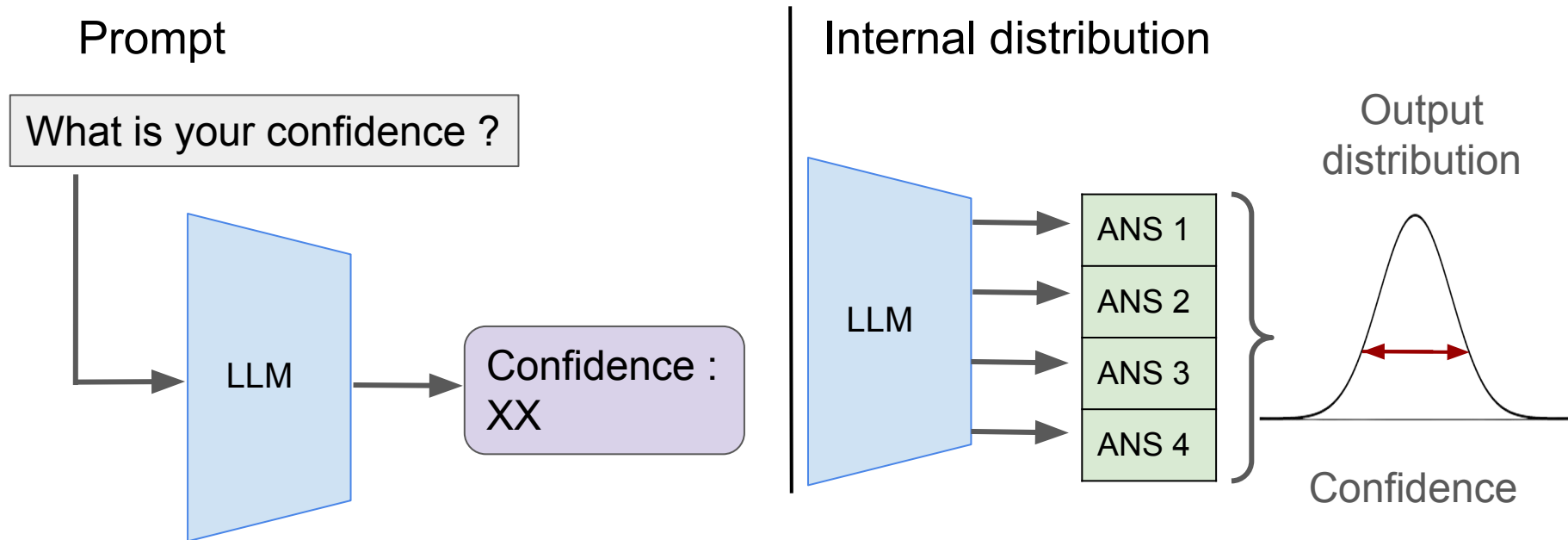
Calibration

Probability of x being correct = confidence score

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$

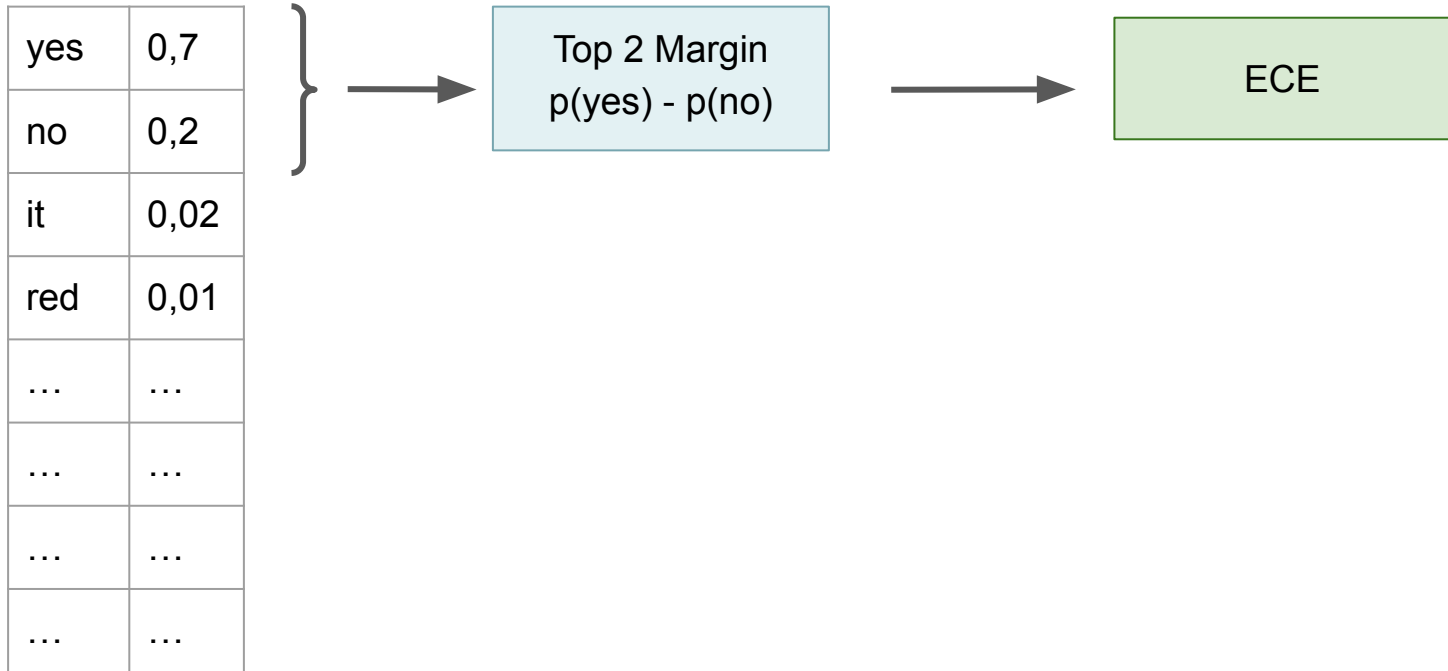


Related Work

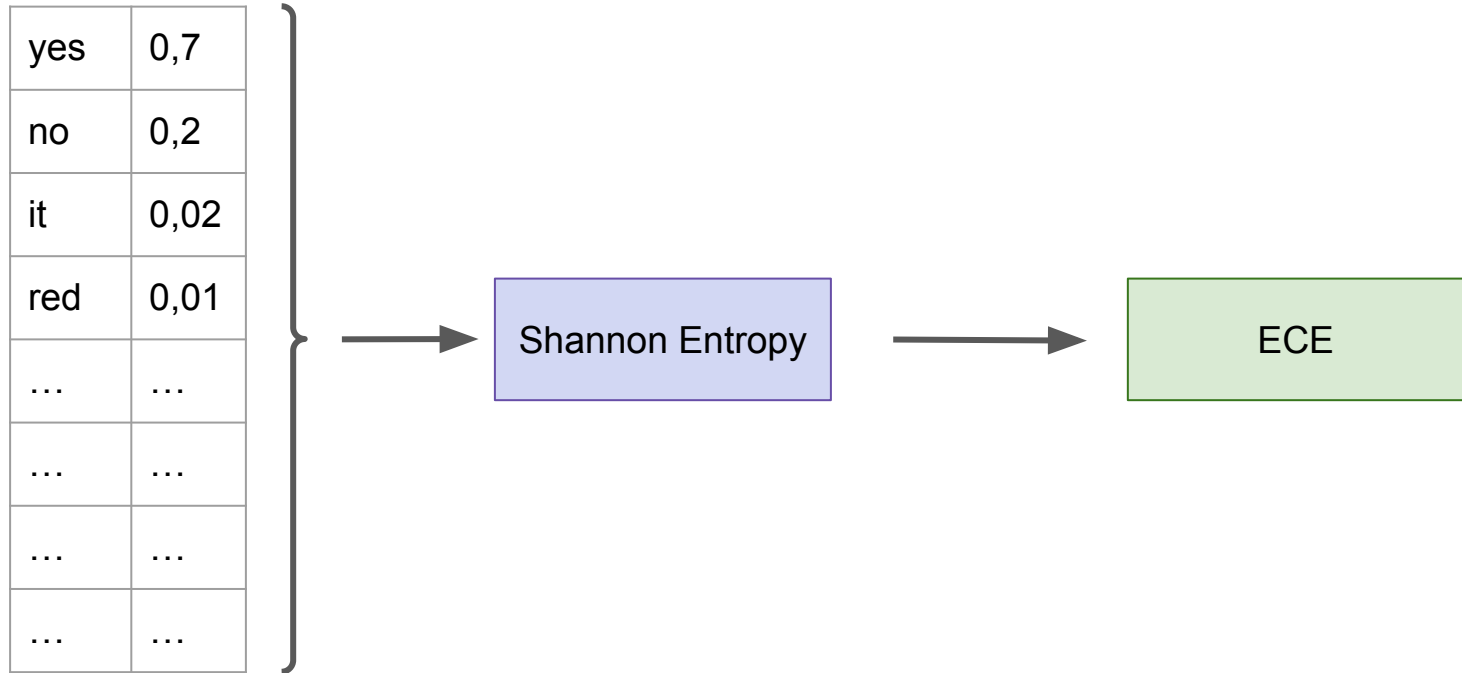


Mostly studied in the LLM domain

Metrics - Internal

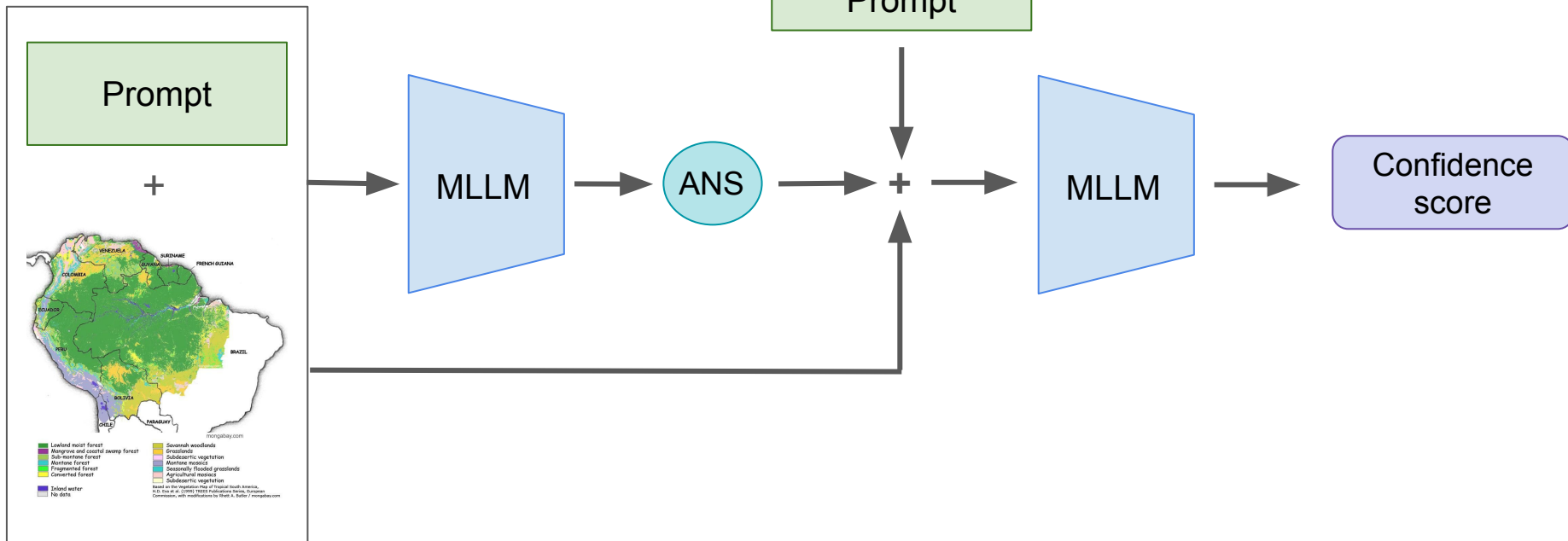


Metrics - Internal



Metrics - External

- Verbalized confidence



Experimental Setup

Models

- Qwen3-VL (2B, 4B, 8B, 32B)
- Qwen3.5 (2B, 4B, 9B, 27B)
- InternVL-3_5 (2B, 4B, 8B, 30B)



Benchmark 1 : HallusionBench

Question : According to the map provided, is New Mexico state larger than Texas state?



Answer : no



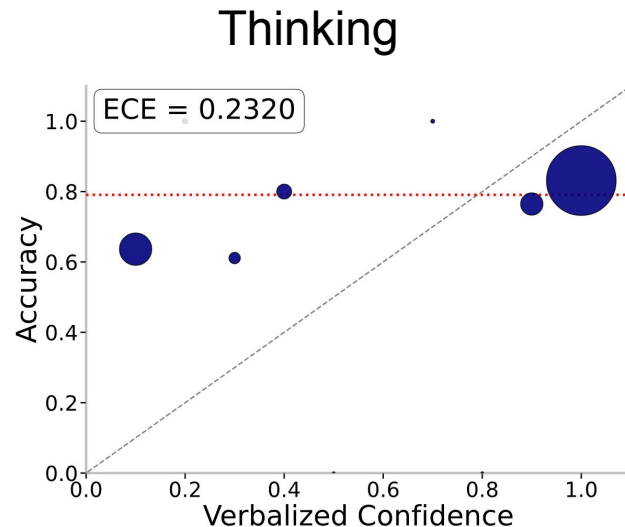
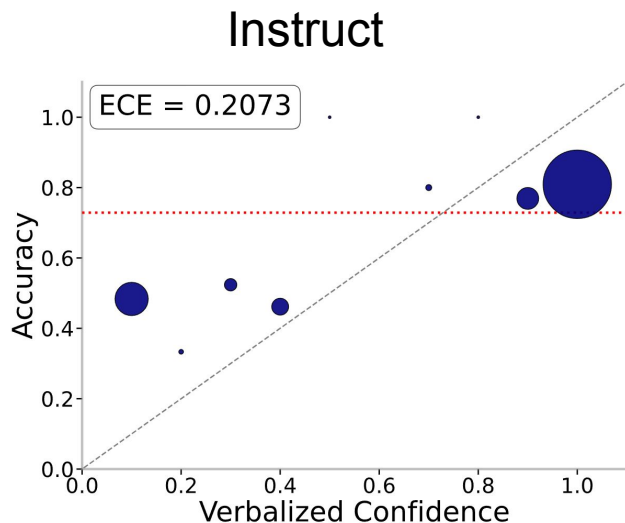
Answer : yes

Benchmark 2 : RefSpatial

Localize objects



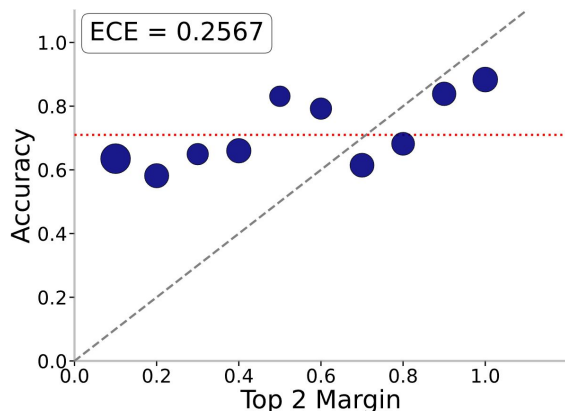
Results : Qwen3.5 9B - HallusionBench



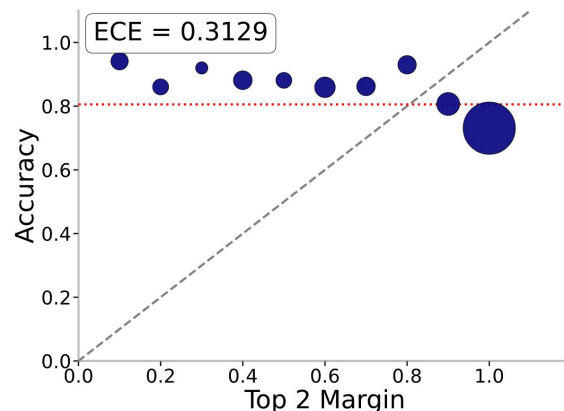
Binary confidence estimation not related to reality
Calibration gets **worse with reasoning**

Results : Qwen3.5 9B - HallusionBench

Instruct



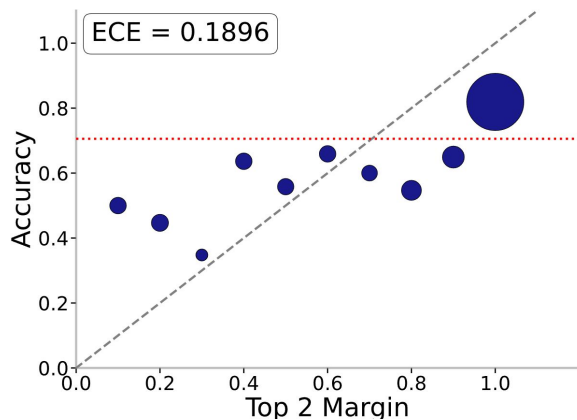
Thinking



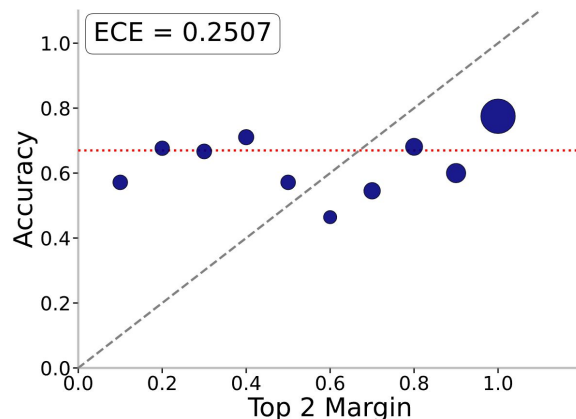
Confidence score **uniformly distributed** across values
Reasoning only adds **overconfidence**

Results : InternVL3_5 8B - Similar results

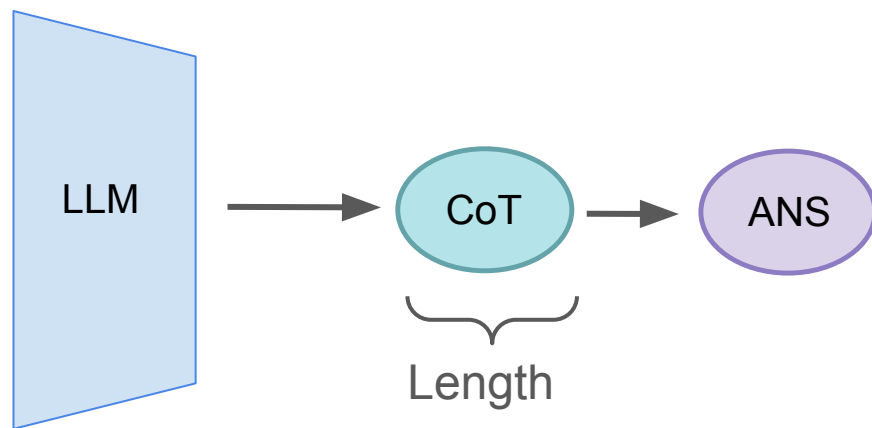
Instruct



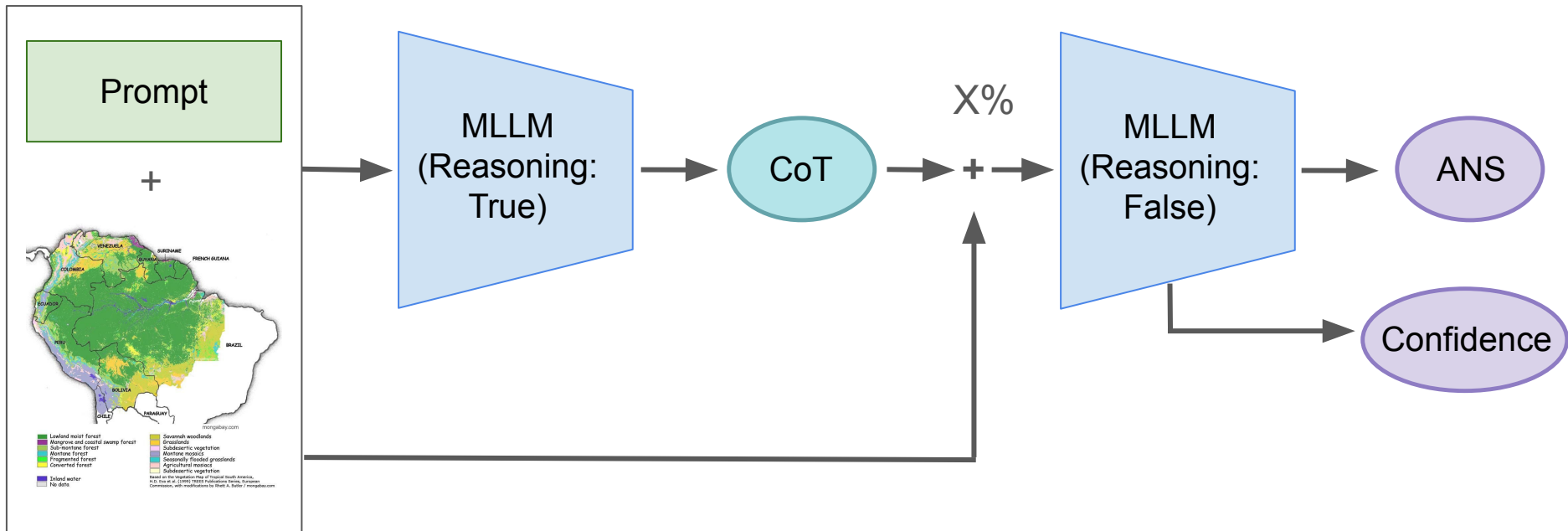
Thinking



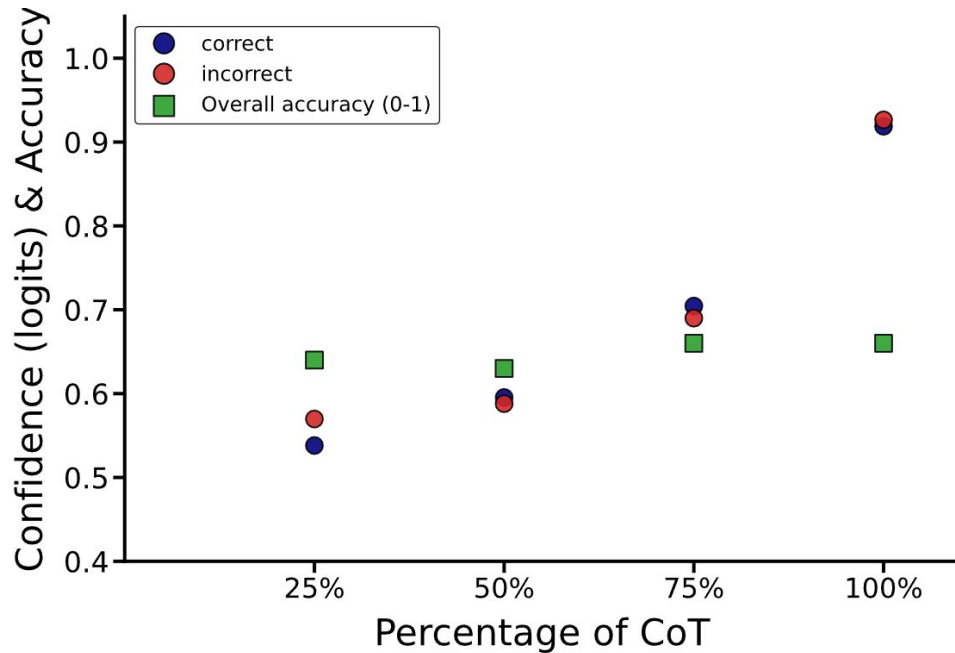
CoT Length as Uncertainty



Track UQ in reasoning



Results : RefSpatial



Reasoning increases
overconfidence

Confidence does not
change with correctness

Conclusion

1. MLLMs **struggle to verbally estimate** their confidence
2. **Internal signals** do **not** reflect better calibration
3. Calibration does **not benefit** from **additional token generation**
4. Reasoning increases **overconfidence**
5. Could we use it to reduce **overthinking** ?

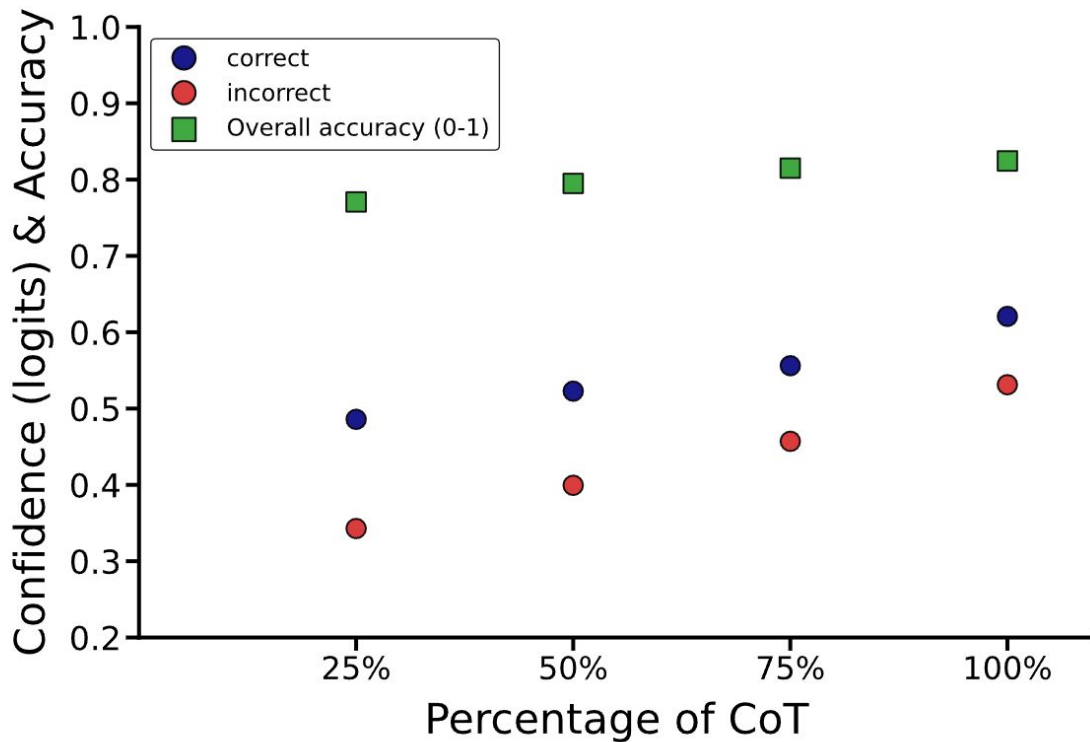
Thank you for your attention!

Questions ?

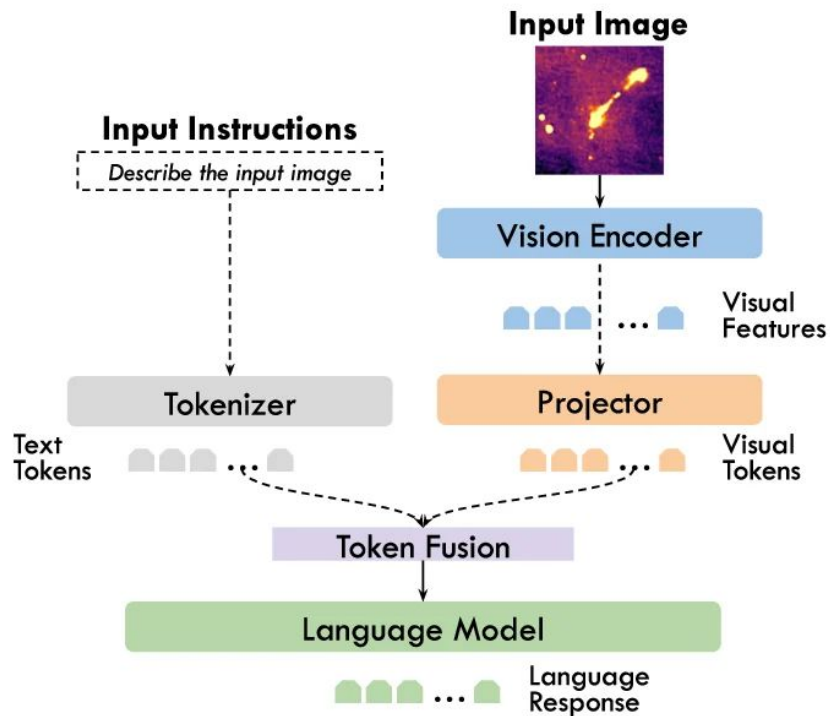
References

- S. Bai et al., Qwen3-VL Technical Report, 2025
- Chuan Guo et al. “On calibration of modern neural networks”. In: ICML 2017
- A. Bradley et al., Trace Length is a Simple Uncertainty Signal in Reasoning Models, 2025

Results : HallusionBench - Track UQ along reasoning



Vision Language Models



Riggi, S. et al., radio-llava: Advancing vision-language models for radio astronomical source analysis, Publications of the Astronomical Society of Australia, 2025

Liu, H., et al., Visual instruction tuning. In *NeurIPS*, 2023

Track UQ in reasoning

The user wants to know if China, Hong Kong SAR is the leading importing country based on the chart.

1. ****Identify the chart type:**** It's a horizontal bar chart.
2. ****Identify the axes:****
 - * Y-axis: Country (China, Hong Kong SAR, France, Germany, Italy, Japan, Singapore, Switzerland,

United Arab Emirates, United Kingdom, United States).

- * X-axis: Import value in million U.S. dollars.
3. ****Locate the relevant data point:**** Look for the bar labeled `"China, Hong Kong SAR"`.
 4. ****Compare with other bars:****
 - * The bar for `"China, Hong Kong SAR"` is the longest bar.
 - * The next longest bar is `"Switzerland"`.
 - * The next is `"United Arab Emirates"`.
 - * The next is `"United States"`.

5. ****Determine the leader:**** Since the bar for China, Hong Kong SAR is the longest, it represents the highest import value.

6. ****Formulate the answer:**** The question asks `"Is China, Hongkong SAR, the leading importing country... with the highest import value in 2018?"`. The answer is yes.

</think>

Yes"

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- </think>

Yes"